

Mohamed Mostafa Hassan Afify

AI Engineer | Generative AI & Agentic Systems | Computer Vision

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PROFESSIONAL SUMMARY

AI Engineer who builds systems, not demos — self-correcting agentic pipelines, multi-agent orchestration, and local vision-language models that run without the cloud. Every project ships with an evaluation layer, because a RAG system that isn't measured is one you can't trust. Skilled in **LangGraph**, **RAG**, **vector search**, **FastAPI** and **computer vision**, with a data-analytics background that keeps model decisions grounded in real data. Trained by **NVIDIA & ITI**, **Google**, **Route Training Center** and **Instant & Orange Digital Center (ODC)**.

EDUCATION

B.Sc. in Computers and Management Information Systems (MIS) 2019 – 2023
El Gazeera Higher Institute, Cairo

PROFESSIONAL EXPERIENCE

Maintenance Engineer — Vision Company, Cairo Dec 2023 – Jan 2025

- Tracked equipment performance and maintenance logs to surface recurring failure patterns, turning scattered records into a usable reporting base.
- Audited inventory and sales data for inconsistencies, resolving data-quality issues that were distorting operational reports.

PROJECTS

Agentic RAG Deep Visualizer [GitHub](#)

- Stateful agentic RAG system (**LangGraph** + **Groq Llama 3.3 70B**) with a self-correcting dual-memory loop: checks a local **Qdrant** store first, then autonomously falls back to live web scraping (**Firecrawl**) and auto-ingests new knowledge.
- Streamlit** dashboard with real-time node-execution tracing and on-the-fly LLM hyperparameter tuning.

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Multi-Agent Deep Researcher [GitHub](#)

- Designed a 4-agent workflow (**Planner**, **Researcher**, **Writer**, **Critic**) on **LangGraph** + **Groq Llama 3.1**; the Critic auto-routes reports back to the Writer for refinement, capped at 2 revisions to control cost.
- Integrated **SerpAPI** for parallel web research; used **Redis** as a persistent state checkpointer for fault-tolerant, multi-user session isolation.

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Multimodal Vision OCR Agent [GitHub](#)

- Built a fully local, cloud-free Vision-Language OCR pipeline using **Qwen2.5-VL (3B)** via **Ollama**, served through a **FastAPI** endpoint.
- Engineered an **OpenCV** preprocessing stage (affine deskewing, CLAHE contrast enhancement, VIT patch-aligned resizing) turning noisy or tilted documents into clean structured JSON.

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RAG Pipeline Evaluation & Quality Observability [GitHub](#)

- Metrics-driven RAG evaluation engine using **Ragas** with **GPT-4o-mini** as LLM-as-a-Judge, auditing a live **Qdrant** + **Groq Llama 3.3** pipeline on Context Precision, Context Recall, Faithfulness and Answer Relevancy.

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GroundedRAG — Safe & Evaluated Document Q&A System [GitHub](#) | [Demo](#)

- Full RAG pipeline (ingestion, embedding, retrieval, generation) with local LLM inference and automated hallucination evaluation.

NVIDIA DLI Generative AI Training — Advanced Level

Real-Time Fire & Smoke AI Guard [GitHub](#) | [Demo](#)

- YOLO detection model reaching **0.739 mAP@0.5** and **86.5% recall**; Streamlit interface for real-time hazard analysis.

Computer Vision Training — Instant & ODC

COVID-19 X-ray Classification [GitHub](#) | [Demo](#)

- MobileNetV2 transfer learning reaching **96.5% accuracy** distinguishing COVID-19 from pneumonia on chest X-rays.

Route AI & Machine Learning Diploma

CERTIFICATIONS & TRAINING

Generative AI Diploma — Route Training Center | 4 Courses + Capstone Projects [View Certificate](#)
NVIDIA DLI — Generative AI Training (Advanced Level) — ITI & NVIDIA | 2 Courses + Capstone
Google AI Professional Certificate — Google / Coursera | 7 Courses [Verify Credential](#)
Computer Vision Training — Instant & Orange Digital Center (ODC) | 2 Projects [View Certificate](#)
NVIDIA DLI — Generative AI Training (Beginner Level) — ITI & NVIDIA | 5 Courses [View Certificate](#)
AI & Machine Learning Diploma — Route Training Center [View Certificate](#)

TECHNICAL SKILLS

- Generative AI & Agents:** LLMs, RAG, LangChain, LangGraph, CrewAI, Agentic Workflows, Prompt Engineering, Vector Search (Qdrant, ChromaDB, Pinecone), Redis, Groq, Ollama, OpenAI API, Gemini API, Hugging Face
- Computer Vision & VLMs:** YOLO, OpenCV, Qwen2.5-VL, Object Detection, Image Classification, Image Processing, OCR (Tesseract), Real-Time Video Analysis, Diffusion Models, CUDA
- Deep Learning & ML:** CNN, Transformers, Transfer Learning, TensorFlow, Scikit-learn, Feature Engineering, Model Evaluation, Classification, Regression
- Evaluation & Observability:** Ragas, LLM-as-a-Judge, Faithfulness / Answer Relevancy / Context Precision / Context Recall, Hallucination Mitigation
- APIs & Development:** Python, FastAPI, Streamlit, Pydantic, Asyncio, Git, Docker, WebSockets
- Data:** SQL Server, Power BI, Tableau, Pandas, NumPy, EDA
- Soft Skills:** Problem Solving, Critical Thinking, Attention to Detail, Research & Analytical Thinking

LANGUAGES

Arabic (Native) • English (Professional Working Proficiency)